PRD — TNBC Drug Response Predictor

Version: 1.0 (MVP for Graduation)

Date: 2026-09-23

Status: Draft

Scope lock: GDSC ranking only / Laptop-Colab / <6 months

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1. One-liner

> An AI system that predicts and ranks GDSC drug sensitivity for Triple-Negative Breast Cancer models from cell-line molecular profiles, validated experimentally on MDA-MB-231 (TNBC) vs MCF7 (ER+ control).

Not a patient-treatment recommender. Not a novel-drug generator.

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2. Problem

Triple-Negative Breast Cancer (TNBC) lacks ER / PR / HER2 targets, so response varies widely across models. Question:

- Can a molecular profile (expression + selected mutations) predict drug sensitivity (AUC) in breast cell models?
- Do top AI-ranked drugs show stronger killing on MDA-MB-231 than MCF7 in vitro (selectivity)?

MCF7 is intentionally ER+ / Luminal — used as non-TNBC control, not as TNBC.

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3. Goals / Non-Goals

Goals (must-have for graduation)

- Build TNBC-aware ML dataset from public data (GDSC + DepMap).
- Train regression model: molecular + drug -> AUC.
- Rank GDSC drugs for MDA-MB-231.
- Explain predictions with SHAP.
- Serve via FastAPI + Streamlit dashboard.
- Wet-lab validate Top 3-5 candidates on MDA-MB-231 vs MCF7.
- Report Predicted vs Experimental correlation.

Non-Goals (explicitly out of MVP)

- Prediction for unseen / novel chemical structures (no drug fingerprints in V1).
- Patient-level / clinical prediction.
- Multi-omics (proteomics, methylation) — V3 only.
- React frontend, Postgres, Docker prod deploy — overkill for MVP.
- Deep Learning as primary model (laptop constraint).

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4. Inputs / Outputs

5.1 Input (MVP)

```
{
  "cell_line_id": "MDA-MB-231",
  "expression": { "TPM vector or precomputed PCA vector": "..." }
}
```

V1 uses expression only. Mutations added in V2 if time allows.

Source of truth for MDA-MB-231 / MCF7 expression: DepMap OmicsExpression (TPM, log2).

5.2 Output

```
{
  "cell_line": "MDA-MB-231",
  "model_version": "catboost-v1.2",
  "ranking": [
    {"rank": 1, "drug": "Drug A", "predicted_AUC": 0.18, "ci_lower": 0.15, "ci_upper": 0.22},
    {"rank": 2, "drug": "Drug B", "predicted_AUC": 0.27}
  ],
  "explanation": {"top_genes": ["gene X", "gene Y"], "shap_plot": "url"}
}
```

Lower AUC = more sensitive. Never display "this drug will cure cancer". Display: predicted sensitivity in cell model.

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5. Functional Requirements

FR-01 Data Ingestion

- Load GDSC2 fitted drug response (AUC, LN_IC50).
- Load DepMap sample_info, OmicsExpression, OmicsMutations (V2).
- Log dataset release versions (e.g. DepMap 24Q4, GDSC2 v8.x).

FR-02 Data Cleaning

- Deduplicate cell_line x drug pairs.
- Unify IDs via Cell Model Passport / COSMIC ID, not name string.
- Handle missing expression (drop genes with >20% missing, median impute rest).
- Log data-quality issues to data_quality_report.md.

FR-03 TNBC Annotation

- Do NOT filter by name only. Use sample_info + literature list for TNBC label.
- Keep columns: cell_line_id, name, cancer_type=breast, subtype=[TNBC, Luminal, HER2+, other].
- Must contain MDA-MB-231 (TNBC) and MCF7 (Luminal control).

FR-04 Dataset Integration

Join key: cell_line_id + drug_id -> one row per pair.

```
ML_ROW = [pca_0..pca_127, drug_one_hot, auc_target, ln_ic50_secondary]
```

FR-05 Training

Support in order:

- DummyMean baseline (must beat this)
- RandomForest
- XGBoost
- CatBoost (expected best for MVP)

CV: GroupKFold by cell_line_id, k=5. Holdout: MDA-MB-231 + MCF7 + 20% TNBC lines fully unseen.

FR-06 Evaluation

Regression: MAE, RMSE, R^2, Spearman.

Ranking (primary for project): Spearman per-cell-line, NDCG@10, Recall@10 for sensitive drugs (AUC < 0.3).

Requirement: report metrics both globally AND on TNBC-only slice.

FR-07 Explainability

SHAP global (top 20 genes) + local (per MDA-MB-231 prediction). Save plots to experiments/.

FR-08 Prediction API

```
POST /predict  -> single drug AUC
POST /rank     -> ranked list for cell_line, top_k param
GET  /health
GET  /model-info -> {model_version, dataset_version, metrics}
```

FR-09 Experimental Logging

Store:

```
drug,     predicted_AUC,     tested_concentrations,     experimental_IC50,     experimental_AUC,
viability_csv_path
```

Comparison figure: predicted vs experimental scatter + Spearman.

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6. Non-Functional Requirements

- Reproducibility: every run logs dataset_version + code_commit + params + seed in MLflow.
- Traceability: prediction response includes model_version.
- Performance: train on Colab CPU <30 min, inference <500ms.
- Privacy: no patient data, no PII.
- Portability: requirements.txt + parquet files, no mandatory DB / Docker for MVP.

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7. Data Architecture

```
data/raw/gdsc/      # GDSC2 fitted CSV
data/raw/depmap/    # expression, sample_info
  ↓
data/silver/        # cleaned parquet
  ↓
data/gold/tnbc_ml_dataset.parquet
```

Validation with Pandera (light): check 0 <= AUC <= 1, no NaN targets.

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8. ML Architecture V1 (locked)

```
Expression (19k genes)
  -> variance filter top 2000
  -> log2(TPM+1) + z-score per gene
  -> PCA 128 (fit on train cell lines only)
  + Drug one-hot (GDSC ~297 drugs)
  ↓
XGBoost / CatBoost regressor -> AUC
```

Why AUC first: bounded [0,1], available in GDSC fitted data, comparable across drugs. LN_IC50 as secondary model only if time.

Why no deep learning: n_samples (~10k breast pairs, ~200k pan-cancer) << n_features, laptop, <6mo.

Future V2: + selected mutations (TP53, BRCA1/2, PIK3CA, PTEN), + CNV, + PRISM external check.

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9. MVP Definition of Done

- [ ] 01_data_audit.ipynb shows intersection counts + MDA-MB-231/MCF7 present
- [ ] tnbc_ml_dataset.parquet built reproducibly
- [ ] CatBoost beats DummyMean on unseen cell lines (R^2>0.25 and per-line Spearman>0.4 on breast slice — threshold to confirm after audit)
- [ ] Ranked Top-20 list for MDA-MB-231 exported
- [ ] SHAP plot + FastAPI + Streamlit working on Colab
- [ ] Wet-lab: dose-response for Top 3-5 + Doxorubicin (pos control) on both lines
- [ ] Final figure: predicted vs experimental

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10. Wet-Lab Validation Protocol (summary)

- Candidates: Top 3-5 AI + 1 positive control (Doxorubicin/Paclitaxel) + 1 negative (predicted resistant).
- Assay: viability (MTT / CellTiter-Glo), 72h, 6-8 concentrations, triplicates.
- Cell lines: MDA-MB-231 and MCF7 side-by-side, same passage window.

- Readout: IC50 + AUC from fitted curve.
- Success: Spearman(predicted_AUC, experimental_AUC) > 0 and selectivity IC50_MCF7 / IC50_MDA231 > 1.5 for at least 1 candidate. This is research signal, not clinical claim.

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11. Risks

Risk | Mitigation
Only ~20 TNBC lines -> overfit | Train pan-cancer/breast, test TNBC slice; PCA + regularization
MDA-MB-231 missing in GDSC/DepMap intersection | Fallback: use CCLE expression + map by name + Passport ID; document version
Drug one-hot can't generalize to new drugs | Accepted; scope locked to GDSC ranking only
Batch effect GDSC vs lab | Use rank order, not absolute AUC; include controls
Colab RAM limits (expression matrix) | Use parquet + float32 + PCA, chunked load

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12. Milestones (20 weeks)

- W1-2: Data audit
- W3-5: Cleaning + gold dataset
- W6-8: Baseline + XGBoost/CatBoost + holdout eval
- W9-10: SHAP + API + Streamlit
- W11-18: Wet-lab (parallel with writing)
- W19-20: Thesis + defense figures

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13. Open Questions

- DepMap release version to freeze? (propose latest 2024+)
- Exact GDSC2 file version?
- Lab max concentrations / available drugs budget?
- Thesis language: EN or AR?

Appendix: Repo Structure

```
TNBC/
├── PRD.md
├── data/raw/ silver/ gold/
├── notebooks/01_data_audit.ipynb
├── src/ingestion/ features/ models/ api/
├── experiments/
├── dashboard/app.py
└── wetlab/results.csv
```